



Armada Kalamunda Group

Intravenous Iron Therapy – Adult Guideline

1. Purpose

This document provides guidance regarding the prescribing, supply, administration and monitoring of intravenous iron products at Armada Kalamunda Group (AKG) sites.

2. Applicability

This guideline is applicable to all Armada Kalamunda clinical staff working in their scope of practice that prescribe, administer, supply, monitor or advise on the use of iron infusions. The guideline applies to patients receiving iron in both inpatient and outpatient settings.

It does not apply to patients receiving iron infusions through dialysis as part of treatment for end stage renal failure. Refer to *Administration of Intravenous Iron Polymaltose for Haemodialysis Patients Procedure*.

3. Background

Parenteral iron administration is indicated to correct iron deficiency states (absolute or functional) associated with anaemia; in situations where oral iron therapy is inappropriate or ineffective, or when a rapid iron supply is required to reduce the need for blood product transfusion.

There are three intravenous iron preparations available for use at AKG sites as per the table below:

Product	Relative cost	Length of infusion	Use at AKG
<i>Ferrosig</i> ® Iron POLYmaltose 100mg/2mL vials	Low cost	Long infusion	Unrestricted
<i>Ferinject</i> ® Ferric CARBOXYmaltose 500mg/10mL	High cost	Short infusion	Restricted ¹
<i>Venofer</i> ® Iron SUCROSE 100mg/5mL	Low cost	Short infusion	Restricted ²

Product	Relative cost	Length of infusion	Use at AKG
¹ Ferric CARBOXYmaltose is restricted for use: ○ In outpatient and day-patient settings. A PBS outpatient prescription is required ○ In patients with a documented hypersensitivity to iron POLYmaltose ○ In inpatients in ED and MAU in situations where a longer infusion would delay discharge. A PBS discharge prescription is required			
² Iron SUCROSE is restricted for use in patients who have a documented hypersensitivity to all other intravenous iron preparations. Iron SUCROSE is not routinely kept by pharmacy at AKG sites.			

4. Guidelines

4.1 Consent

- When obtaining patient consent for the administration of intravenous iron therapy clinicians must adhere to the principles of the [WA DoH Consent to Treatment Policy OD 0657/16](#).
- Document the consent on the AKG [Iron Infusion Consent and Checklist \(AKMR30.11\)](#) form to be filed in the patient's medical record.
- Ensure patient is provided with [written information](#) relating to intravenous iron infusions.

Alerts

Subcutaneous leakage of iron - iron infusions that leak under the skin result in permanent discolouration of the skin and can happen in the absence of patient reports of discomfort, while the risk is low patients should be informed of this possibility prior to infusion.

4.2 Adverse effects

Fatal anaphylactic and anaphylactoid reactions have occurred, including in patients who have tolerated previous infusions and test doses (which are therefore no longer recommended). These reactions are most common in the first few minutes of the infusion and are characterised by sudden onset of difficulty breathing, tachycardia and hypotension. The risk is higher in patients with inflammatory and autoimmune conditions and allergies.

Risk mitigation strategies include:

- Ensure patient is provided with clear verbal information relating to possible adverse reactions (immediate and delayed) and to inform nursing staff if any signs of headache, joint/muscle pain, pain at the site of injection site, skin staining, dizziness, nausea and vomiting or tachycardia throughout transfusion.
- A volume-controlled device is to be used in the administration of IV iron medication
- Infusions should not be administered after hours where possible. A medical officer should be readily contactable throughout the infusion and notified in the event of any adverse reaction. Moderate to severe reactions warrant a code blue.
- Resuscitation kits and oxygen should be readily available
- Patients should be advised to promptly report any adverse effects

- Patients should be observed for the first 5 minutes of the infusion
- Patients should remain in hospital for a minimum of 30 minutes post-infusion
- If cubital fossa is used for transfusion, splint the site to decrease chance of extravasation
- Observations should be documented on the Adult Observation Chart (AKMR 140.3) at the intervals indicated below:

Product	Observation interval
Iron POLYmaltose	At baseline and every 5 minutes for the first 15 minutes, then at 30minute intervals (or 15minute intervals if the rapid protocol is used) for the duration of the infusion, and 30 minutes post-infusion.
Ferric CARBOXYmaltose	At baseline and at completion of infusion

4.3 Prescribing and administration

- The need for intravenous iron is to be determined by a Medical Officer after reviewing the relevant products contraindications and precautions and should have Consultant approval.
- Intravenous iron orders are to be prescribed on an Intravenous Fluid Therapy and Consent for Transfusion form (AKMR176) – for ferric CARBOXYmaltose, a PBS prescription is also required.
- Calculation of iron dosing is complex and depends on whether the intention is to treat anaemia or fully replace iron deficit. The dosing information below is a guide only. Higher doses may be required for patients with severe anaemia or to fully replace total iron deficit. Total iron deficit can be calculated using the Gazoni formula or the Simplified Method. Please review the product information or seek haematology advice for further information.
- Where possible, oral iron supplements should be ceased one week prior to iron infusions and should not be recommenced for one week following infusions.

Alerts

- Infusion rates, maximum doses and dilutions are not interchangeable between intravenous iron preparations
- For specific patient populations (e.g. obstetrics, perioperative or dialysis patients) please seek specialist advice.

Iron POLYmaltose (Ferrosig®)

Dosing:

- <70kg IBW: 1000mg elemental iron (10 vials)
- >70kg IBW: 1500mg elemental iron (15 vials)

Unless otherwise indicated by body size and severity of anaemia, the usual maximum dose is 1500mg. The dose should not exceed 2500mg in a single infusion.

Slow Protocol Rate

Add dose (up to 1500mg) to 250mL of normal saline.

Commence infusion at 40mL/hour. If no reaction occurs within the first 1 hour, increase the infusion rate to 120mL/hour.

Rapid Protocol Rate

Patients receiving the rapid protocol must

- Be haemodynamically stable and otherwise well
- Not have experienced a previous reaction to an iron infusion
- Not have an autoimmune or inflammatory disease state
- Not be in cardiac failure

Add dose (up to 1500mg) in 250mL of normal saline.

Commence infusion at 40mL/hour for 15 minutes. If tolerated, increase the rate to 250mL/hour.

*IBW = ideal body weight

Ferric CARBOXYmaltose (Ferinject®)

Dosing and administration:

- <70kg IBW: 500mg (10mL) in 100mL normal saline over 15 minutes
- >70kg IBW: 1000mg (20mL) in 250mL normal saline over 15 minutes

The maximum single dose is 1000mg. For calculated doses >1000mg, give the remainder of the dose at least 7 days after the initial infusion.

*Two 500mg doses can be administered in the same week providing they are at least 48 hours apart

*IBW= Ideal body weight

4.4 Extravasation

The patient and nurse should be alert to extravasation as this may result in pain, necrosis and irreversible skin staining. To reduce risk, ensure the cannula is in the largest vein possible, secured and immobilised. Administration should be via the cubital fossa for all except renal patients. Flush the line with 50mL sodium chloride before and after infusion to reduce risk.

4.4.1 Treatment of extravasation

- Stop the infusion, aspirate any residual iron from the cannula and then remove the cannula
- Do not attempt to re-site
- Minimise movement and avoid massage of the area to reduce spread
- Reassure and provide a full explanation to the patient
- Assess the volume of infiltration by recording the volume of infused fluid
- Inform the RMO for assessment of the patient
- Apply hydrocortisone cream to the affected site and elevate the limb
- If iron staining is visible, measure the site and arrange for hospital photographs to be taken
- Enter as a clinical incident on the AKG Datix system and document in the patient's medical record
- Refer the patient to their General Practitioner for follow up of the progress of any extravasation.

4.5 Management of adverse reactions

- Immediately stop the infusion
- Inform shift coordinator and request review by medical officer
- Inform the on call obstetric registrar if patient is pregnant
- Perform vital signs as indicated by clinical condition
- Activate medical emergency as per criteria
- Document on Adult Observation and Response Chart (AORC) or Maternity Adult Deterioration Chart (MADDS) and in the patient's integrated notes

4.6 Post infusion follow-up

- Iron studies are not reliable within two weeks of infusion. Haemoglobin and iron stores may be reassessed 21–28 days after infusion.
- Inform patients of possible delayed adverse reactions to iron infusions and advise them to contact their GP or Emergency Department if they become unwell.
- To minimise delayed reactions, advise patients to avoid excess physical activity for 48 hours after infusion.

5. Legislative context

The following documents are required to give effect to this policy [i.e. the documents included are mandatory]:

- [WA Health Consent to Treatment Policy](#)
- [EMHS Consent Policy](#)

6. Supporting information

The following documents support the implementation of this policy:

- [AKG Medication Management](#)

- [AKG Administration of Intravenous Iron Polymaltose for Haemodialysis Patients Procedure](#)

7. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated by the AKG Blood Committee.

Performance measures include:

- Percentage of cases that had a completed consent form at time of non-RBC product transfusion
- Review of Clinical Incidents related to extravasion of Iron Infusions

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.1 Clinicians use the safety and quality Systems form the Clinical Governance Standard when:
 - a. Implementing policies and procedures for medication management
 - b. Managing risks associated with medication management
 - c. Identifying training requirements for medication management
- Standard 7 – Action 7.10 The health service organisation has processes to:
 - a. Manage the availability of blood and blood products to meet clinical need
 - b. Eliminate avoidable wastage
 - c. Respond in times of shortage

9. References

1. Burridge, N. and Symons, K. (2018). Australian Injectable Drugs Handbook. Collingwood: Society of Hospital Pharmacists of Australia.
2. Ferrosig Product Information (2018) In MIMS Online. Retrieved from <http://www.mims.com.au>
3. Ferinject Product Information (2018) In MIMS Online. Retrieved from <http://www.mims.com.au>
4. Venofer Product Information (2018) In MIMS Online. Retrieved from <http://www.mims.com.au>

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
Absolute iron deficiency	This occurs when there is no /insufficient iron available for haemoglobin production
Functional iron deficiency	This is a failure to release iron rapidly enough to meet the bone marrow demand for RBC production, despite adequate iron stores.
Extravasation	The seepage of a fluid from a blood vessel into the tissues

11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	29/05/2023	29/05/2026	Review following recommendation from a SAC 1 investigation

12. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Alison Parr – Director Clinical Services
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